KAIZEN MATHS

Lesson starter

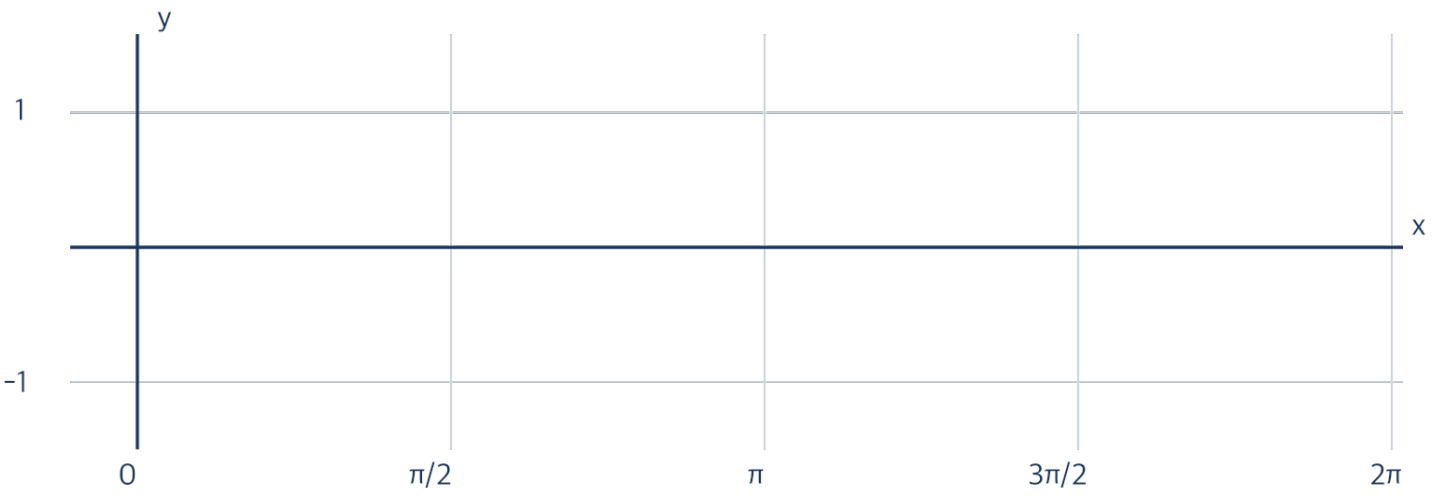
Trig Identities, Proofs and Equations

Graphs & Reciprocals

Name _____

SA1

Sketch the graphs of $\sin x$ and $\operatorname{cosec} x$ on the same set of axes for $0 \leq x \leq 2\pi$.



Show your working

KAIZEN MATHS

Lesson starter

Trig Identities, Proofs and Equations

Graphs & Reciprocals

Name _____

SB1

Rewrite $\sec x$ using sine, cosine, tangent, or a standard identity.

Show your working

KAIZEN MATHS

Lesson starter

Trig Identities, Proofs and Equations

Graphs & Reciprocals

Name

SC1

Solve $2\,\text{cosec}\,\theta = -5$ for $0 \leq \theta \leq 360°$. Give answers to 1 decimal place.

Show your working

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Lesson starter

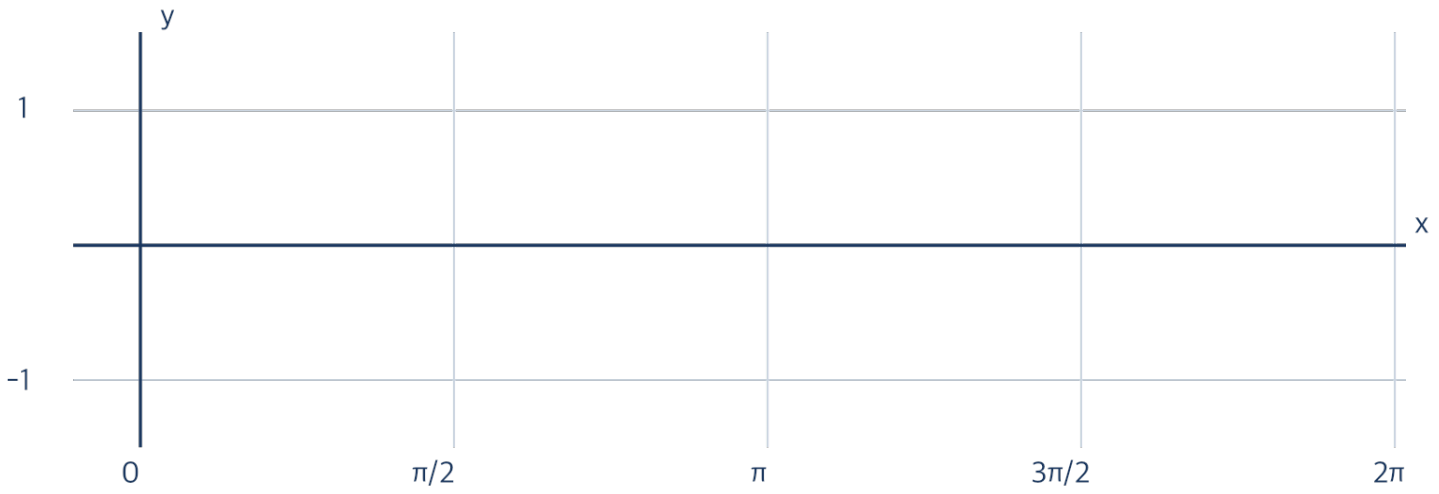
Trig Identities, Proofs and Equations

Graphs & Reciprocals

Name

SD1

Sketch the graphs of $\cos x$ and $\sec x$ on the same set of axes for $0 \leq x \leq 2\pi$.



Show your working

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Exit ticket

Trig Identities, Proofs and Equations

Graphs & Reciprocals

Name

EA1

Rewrite $\tan^2 x$ using sine, cosine, tangent, or a standard identity.

Show your working

KAIZEN MATHS

Exit ticket

Trig Identities, Proofs and Equations

Graphs & Reciprocals

Name

EB1

Solve $3\sec\theta = 8$ for $0 \leq \theta \leq 360°$. Give answers to 1 decimal place.

Show your working

KAIZEN MATHS

Exit ticket

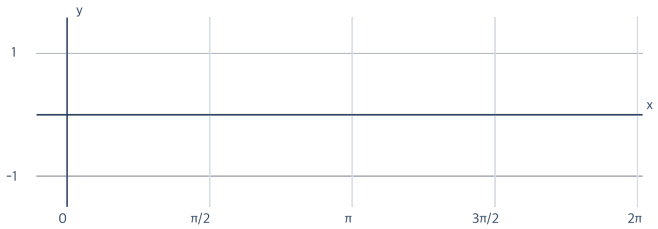
Trig Identities, Proofs and Equations

Graphs & Reciprocals

Name

EC1

Sketch the graphs of $\tan x$ and $\cot x$ on the same set of axes for $0 \leq x \leq 2\pi$.



Show your working

KAIZEN MATHS

Exit ticket

Trig Identities, Proofs and Equations

Graphs & Reciprocals

Name

EA1

Rewrite $\tan^2 x$ using sine, cosine, tangent, or a standard identity.

Show your working

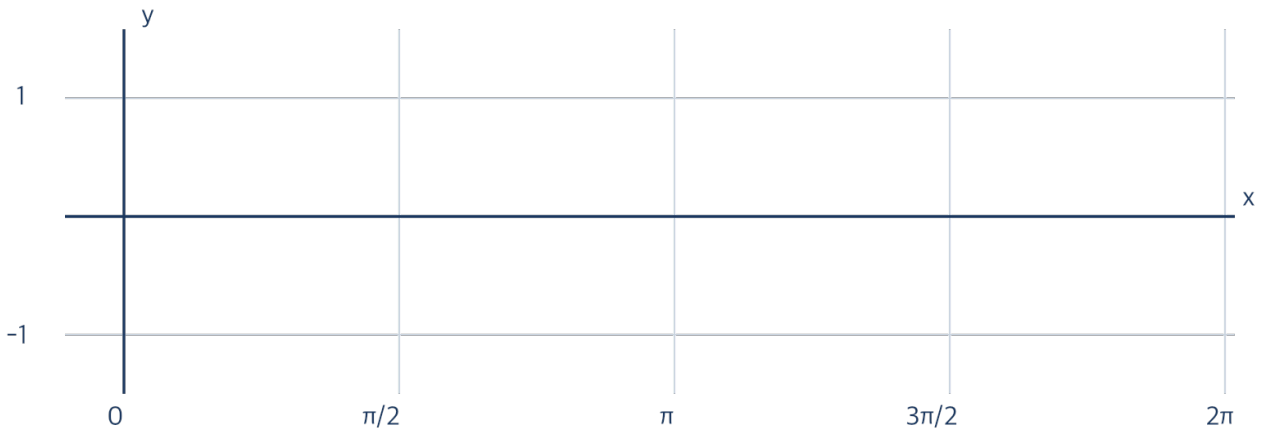
Teacher answers & working

Trig Identities, Proofs and Equations · Graphs & Reciprocals

ALG-44E5-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

Sketch the graphs of $\sin x$ and $\operatorname{cosec} x$ on the same set of axes for $0 \leq x \leq 2\pi$.



Answer:

Key features

Working:

Sketch the base graph $\sin x$ first. Use the standard points $(\frac{\pi}{2}, 1)$, $(\frac{3\pi}{2}, -1)$.

The reciprocal graph is not defined where the base graph is zero. Draw vertical asymptotes at $x = 0, \pi, 2\pi$.

Where the base graph has value 1 or -1, the reciprocal graph has the same value. Use those shared points to place the branches.

The sketch should show the base graph, the reciprocal branches, and the correct vertical asymptotes.

SB1

Rewrite $\sec x$ using sine, cosine, tangent, or a standard identity.

Answer:

$$\frac{1}{\cos x}$$

Working:

Identify the reciprocal or identity needed.

Here, sec is the reciprocal of cos.

$$\sec x = \frac{1}{\cos x}$$

Teacher answers & working

Trig Identities, Proofs and Equations · Graphs & Reciprocals

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SC1

Solve $2\,\text{cosec}\,\theta = -5$ for $0 \leq \theta \leq 360°$. Give answers to 1 decimal place.

Answer:

$\theta = 203.6°, 336.4°$

Working:

Divide by the coefficient.

$\text{cosec}\,\theta = -\frac{5}{2}$

Use the reciprocal definition.

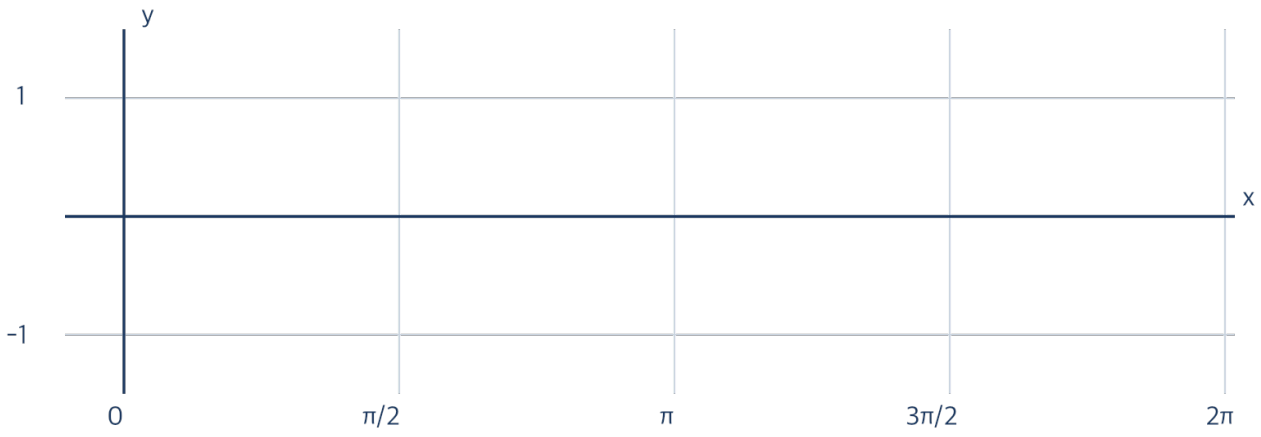
$sin\,\theta = -0.4000$

Find every solution in the interval $0 \leq \theta \leq 360°$.

$\theta = 203.6°, 336.4°$

SD1

Sketch the graphs of $\cos x$ and $\sec x$ on the same set of axes for $0 \leq x \leq 2\pi$.



Answer:

Key features

Working:

Sketch the base graph $\cos x$ first. Use the standard points $(0, 1)$, $(\pi, -1)$, $(2\pi, 1)$.

The reciprocal graph is not defined where the base graph is zero. Draw vertical asymptotes at $x = \frac{\pi}{2}, \frac{3\pi}{2}$.

Where the base graph has value 1 or -1, the reciprocal graph has the same value. Use those shared points to place the branches.

The sketch should show the base graph, the reciprocal branches, and the correct vertical asymptotes.

Teacher answers & working

Trig Identities, Proofs and Equations · Graphs & Reciprocals

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EA1

Rewrite $\tan^2 x$ using sine, cosine, tangent, or a standard identity.

Answer:

$\sec^2 x - 1$

Working:

Identify the reciprocal or identity needed.

Here, from $1 + \tan^2 x = \sec^2 x$.

$\tan^2 x = \sec^2 x - 1$

EB1

Solve $3 \sec \theta = 8$ for $0 \leq \theta \leq 360°$. Give answers to 1 decimal place.

Answer:

$\theta = 68.0°, 292.0°$

Working:

Divide by the coefficient.

$\sec \theta = \frac{8}{3}$

Use the reciprocal definition.

$cos\, \theta = 0.3750$

Find every solution in the interval $0 \leq \theta \leq 360°$.

$\theta = 68.0°, 292.0°$

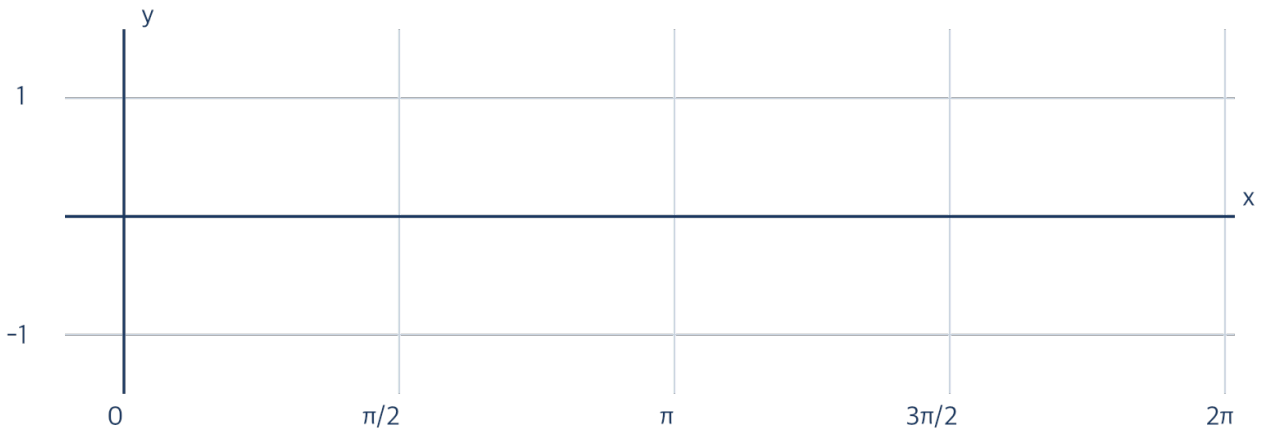
Teacher answers & working

Trig Identities, Proofs and Equations · Graphs & Reciprocals

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EC1

Sketch the graphs of $\tan x$ and $\cot x$ on the same set of axes for $0 \leq x \leq 2\pi$.



Answer:

Key features

Working:

Sketch the base graph $\tan x$ first. Use the standard points $\tan x$ passes through $(0, 0)$, $(\pi, 0)$, $(2\pi, 0)$.

The reciprocal graph is not defined where the base graph is zero. Draw vertical asymptotes at $x = \frac{\pi}{2}, \frac{3\pi}{2}$ for $\tan$; $0, \pi, 2\pi$ for $\cot$.

Where the base graph has value 1 or -1, the reciprocal graph has the same value. Use those shared points to place the branches.

The sketch should show the base graph, the reciprocal branches, and the correct vertical asymptotes.